

A sharp L_q/L_4 inequality for every even q A partial solution of Conjecture 16 in arXiv:2110.14382

Discovery packet for human verification

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Status. `partial_solution_likely_valid`. The argument below proves the inequality part of Eitan's conjecture for the infinite family $p = 4$, $q \in 2\mathbb{N}$, $q > 4$. It does not settle the other values of p .

1 Source question and result

Let $\mathcal{E}$ be a standard exponential random variable and put $\Gamma = \mathcal{E} - 1$. Eitan's Conjecture 16 asks whether, for every centered real log-concave random variable X and all even integers $p < q$,

$$\frac{\|X\|_q}{\|X\|_p} \leq \frac{\|\Gamma\|_q}{\|\Gamma\|_p},$$

with the stated equality characterization [1, Conjecture 16, p. 10]. Here $\mathbb{E}\Gamma^n = !n$, the subfactorial of n .

Calculating the coefficients of the first few polynomials we get:

$$\begin{aligned}\tilde{h}_4(s) &= -96 \\ \tilde{h}_6(s) &= -720(15 + 8s^2 + 144s^4) \\ \tilde{h}_8(s) &= -1680(1485 - 2880s^2 + 105696s^4 + 104448s^6 + 268544s^8) \\ \tilde{h}_{10}(s) &= -5040(269325 - 1323000s^2 + 72560880s^4 + 280339200s^6 \\ &\quad + 1409629440s^8 + 1162622976s^{10} + 1050406912s^{12}).\end{aligned}$$

One can easily check that when $q \leq 10$, $a_{i,q} \leq 0$ for all $i \neq 1$ and $a_1^2 - 4a_0a_2 < 0$. Thus $\tilde{h}_q$ never changes sign and $\tilde{h}_q(s)$ has the same sign as $s(s-1)(s+1)$, which proves the theorem. Using computer assistance one can see the same proof still holds for $q < 100$. For details see [5]. $\square$

We finish this paper with a conjecture which generalizes these empiric results:

Conjecture 16. *Let X be a centered log-concave random variable. Then for even integers $p < q$, we have:*

$$(3.15) \quad \frac{\|X\|_q}{\|X\|_p} \leq \frac{\|\Gamma\|_q}{\|\Gamma\|_p} = \frac{(!q)^{\frac{1}{q}}}{(!p)^{\frac{1}{p}}},$$

with equality if and only if X coincides with $\pm\Gamma$ in law, where Γ is a random variable with density $g(x) = e^{-(x+1)}1_{[-1,\infty)}$.

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Theorem 1 (The $p = 4$ anchored family). *For every centered real log-concave random variable X and every integer $m \geq 3$,*

$$\frac{\|X\|_{2m}}{\|X\|_4} \leq \frac{\|\Gamma\|_{2m}}{\|\Gamma\|_4} = \frac{(!2m)^{1/(2m)}}{9^{1/4}}.$$

The bound is sharp. Within Eitan's one-parameter extremal family, equality holds only at the two endpoint laws Γ and $-\Gamma$.

2 Idea of the proof

Eitan's localization theorem reduces the sharp log-concave comparison to a one-parameter family made from two independent centered exponentials. After the symmetric coordinate $y = z^2$ is introduced, all even moments become polynomials in y . For $p = 4$, the desired logarithmic derivative has a factor $1 - y$. The point is to prove that the quotient by $1 - y$ has strictly positive coefficients at every order.

A direct expansion leaves alternating sums. The useful extra substitution is $w = v/(1 + v)$. It turns the coefficient in question into a convolution with $(1 - w)^{-2n}$, hence with a positive kernel. All coefficients from degree four onward are controlled by two elementary series whose coefficients satisfy positive first-order recurrences. The four exceptional coefficients are not individually positive, but their entire contribution collapses to an explicit positive cubic. This is an all-orders certificate, not a finite computer check.

3 Reduction to moment polynomials

Let Γ' be an independent copy of Γ . Eitan's Theorem 12 gives, for $1 < p < q$,

$$\frac{\|X\|_q}{\|X\|_p} \leq \max_{0 \leq s \leq 1} \frac{\|\Gamma^s\|_q}{\|\Gamma^s\|_p}, \quad \Gamma^s = s\Gamma - (1 - s)\Gamma'.$$

Set

$$z = 1 - 2s, \quad a = \frac{1 - z}{2}, \quad b = \frac{1 + z}{2}, \quad X_z = a\Gamma - b\Gamma'.$$

Reflection permits us to take $0 \leq z \leq 1$. Independence gives the moment generating function

$$M_z(t) = \mathbb{E}e^{tX_z} = \frac{e^{zt}}{(1 - at)(1 + bt)} = \frac{e^{zt}}{1 + zt - (1 - z^2)t^2/4}. \quad (1)$$

Put $y = z^2$,

$$r_{2m}(y) = \mathbb{E}X_z^{2m}, \quad e_m(y) = \frac{r_{2m}(y)}{(2m)!}, \quad g(y) = y^2 + 4y + 1.$$

Equation (1) shows that e_m is a polynomial of degree m . Also

$$r_4(y) = \frac{3}{2}g(y). \quad (2)$$

For $m \geq 3$, monotonicity of $\|X_z\|_{2m} / \|X_z\|_4$ is therefore equivalent to

$$L_m(y) := 2g(y)e'_m(y) - mg'(y)e_m(y) \geq 0 \quad (0 \leq y \leq 1). \quad (3)$$

4 The coefficient certificate

Lemma 1. *For every $m \geq 3$, there are strictly positive numbers $\rho_{m,0}, \dots, \rho_{m,m-1}$ such that*

$$L_m(y) = (1 - y) \sum_{j=0}^{m-1} \rho_{m,j} y^j. \quad (4)$$

Proof. Write

$$4^m e_m(y) = \sum_{k=0}^m A_{m,k} y^k.$$

Partial fractions in (1), followed by coefficient extraction, give

$$A_{m,k} = \sum_{\ell=0}^{2k} \frac{(-1)^\ell 2^\ell}{\ell!} \binom{2m-\ell+1}{2k-\ell} \quad (5)$$

$$= [v^{2k}] (1+v)^{2m+1} \exp\left(-\frac{2v}{1+v}\right). \quad (6)$$

For completeness, the first line follows by writing the coefficient of t^{2m} in

$$\frac{e^{zt}}{(1-at)(1+bt)} = a \frac{e^{zt}}{1-at} + b \frac{e^{zt}}{1+bt}$$

and then collecting the coefficient of z^{2k} ; the second line is the binomial theorem.

Let

$$\Phi_m(v) = (1+v)^{2m+1} \exp\left(-\frac{2v}{1+v}\right).$$

The even part of $\Phi_m(v)$ is $4^m e_m(v^2)$. Odd terms do not contribute to the even coefficient below. From (3),

$$4^m \rho_{m,j} = [v^{2j}] \frac{(1+4v^2+v^4)\Phi'_m(v)/v - m(4+2v^2)\Phi_m(v)}{1-v^2}. \quad (7)$$

Since

$$\frac{\Phi'_m(v)}{\Phi_m(v)} = \frac{(2m-1) + (2m+1)v}{(1+v)^2},$$

the rational expression in (7) simplifies to

$$4^m \rho_{m,j} = [v^{2j+1}] (1+v)^{2m-2} e^{-2v/(1+v)} (2m(v^3+1) - v^4 - 4v^2 - 1). \quad (8)$$

We first dispose of $j = 0, 1$, directly from (5):

$$4^m \rho_{m,0} = 2(m-2)(2m-1), \quad (9)$$

$$4^m \rho_{m,1} = \frac{2(m-2)}{3} (4m^3 - 24m^2 + 47m - 31). \quad (10)$$

The first quantity is positive. If $h = m-3 \geq 0$, the cubic in (10) is

$$4h^3 + 12h^2 + 11h + 2 > 0.$$

Fix now $2 \leq j \leq m-1$, and write

$$n = m-j-1 \geq 0, \quad N = 2j+1.$$

Use $w = v/(1+v)$ and the elementary coefficient identity

$$[v^N] H(v) = [w^N] (1-w)^{N-1} H\left(\frac{w}{1-w}\right).$$

Equation (8) becomes

$$4^m \rho_{m,j} = [w^N](1-w)^{-2n} B(w) A_m(w), \quad (11)$$

where

$$\begin{aligned} B(w) &= \frac{e^{-2w}}{(1-w)^4}, \\ A_m(w) &= (2m-1) + (-8m+4)w + (12m-10)w^2 \\ &\quad + (-6m+12)w^3 - 6w^4. \end{aligned}$$

Introduce

$$\begin{aligned} A_1(w) &= 2 - 8w + 12w^2 - 6w^3, \\ C(w) &= 1 - 4w + 2w^2 + 6w^3 - 6w^4, \\ U(w) &= B(w)A_1(w), \quad D_j(w) = B(w)(jA_1(w) + C(w)). \end{aligned}$$

As $m-1 = j+n$, (11) is

$$4^m \rho_{m,j} = [w^N](1-w)^{-2n} (D_j(w) + nU(w)). \quad (12)$$

We show that all coefficients of $D_j + nU$ from degree four through degree N are positive. Let $B(w) = \sum_{k \geq 0} b_k w^k$. The differential identity

$$(1-w)B'(w) = 2(1+w)B(w)$$

gives

$$(k+1)b_{k+1} = (k+2)b_k + 2b_{k-1}.$$

Thus $b_k > 0$, $b_{k+1} > b_k$, and, for $k \geq 2$, $b_{k+1} < 2b_k$ (the last inequality is equivalent to $2b_{k-1} < kb_k$).

Write $U(w) = \sum u_k w^k$. Since

$$U(w) = 2(3w^2 - 3w + 1) \frac{e^{-2w}}{(1-w)^3},$$

and the coefficients a_k of $e^{-2w}/(1-w)^3$ obey

$$a_{k+1} = a_k + \frac{2a_{k-1}}{k+1},$$

they are positive and nondecreasing. For $k \geq 4$,

$$\begin{aligned} \frac{u_k}{2} &= a_k - 3a_{k-1} + 3a_{k-2} \\ &= \left(1 + \frac{2}{k}\right) a_{k-2} - \frac{4}{k-1} a_{k-3} > 0, \end{aligned}$$

because $a_{k-2} \geq a_{k-3} > 0$ and $1 + 2/k > 4/(k-1)$.

If $c_k = [w^k]B(w)C(w)$, direct differentiation gives the useful identity

$$\sum_{k \geq 0} \left(\frac{k-1}{2} u_k + c_k \right) w^k = \frac{w}{2} U'(w) - \frac{1}{2} U(w) + B(w)C(w) = 2w(2w-1)B(w). \quad (13)$$

Its k -th coefficient is $4b_{k-2} - 2b_{k-1} > 0$ for $k \geq 4$. Whenever $4 \leq k \leq N = 2j + 1$, we have $j \geq (k - 1)/2$, so

$$[w^k]D_j = ju_k + c_k \geq \frac{k-1}{2}u_k + c_k > 0.$$

Adding nU preserves strict positivity.

Only degrees zero through three remain. Their coefficients in $D_j + nU$ are, using $m = j + n + 1$,

$$2m - 1, \quad -2(2m - 1), \quad 4m - 6, \quad -\frac{2(m - 2)}{3}. \quad (14)$$

If $n = 0$, these degrees cannot contribute to $[w^N]$, since $N \geq 5$, and the already positive degree- N coefficient finishes the proof. Suppose $n \geq 1$, and write

$$K_r = [w^r](1 - w)^{-2n} = \binom{2n + r - 1}{r}.$$

The total contribution of (14) is

$$S = (2m - 1)K_N - 2(2m - 1)K_{N-1} + (4m - 6)K_{N-2} - \frac{2(m - 2)}{3}K_{N-3}.$$

Using $K_r/K_{r-1} = (2n + r - 1)/r$, this becomes

$$S = K_{N-3} \frac{2(j + n - 1)P(j, n)}{3j(2j - 1)(2j + 1)}, \quad (15)$$

where

$$\begin{aligned} P(j, n) &= 8j^3 + 12j^2n - 12j^2 + 12jn^2 - 12jn - 2j \\ &\quad + 12n^3 - 12n^2 - 3n + 3. \end{aligned}$$

Putting $h = n - 1 \geq 0$ makes positivity manifest:

$$\begin{aligned} P(j, h + 1) &= 8j^3 - 2j + 12j^2h + 12j(h^2 + h) \\ &\quad + 12h^3 + 24h^2 + 9h > 0 \end{aligned}$$

for $j \geq 2$. Hence $S > 0$. The tail of the convolution in (12) is also positive, proving $\rho_{m,j} > 0$. Together with (9)–(10), this proves the lemma. $\square$

Proof of Theorem 1. Lemma 1 gives $L_m(y) > 0$ for $0 \leq y < 1$ and $L_m(1) = 0$. Thus

$$\frac{d}{dy} \log \frac{r_{2m}(y)^2}{r_4(y)^m} = \frac{2r'_{2m}(y)}{r_{2m}(y)} - \frac{mr'_4(y)}{r_4(y)} \geq 0.$$

Consequently $\|X_z\|_{2m} / \|X_z\|_4$ is maximized at $y = 1$, whose laws are the two centered-exponential endpoints under reflection. Eitan's Theorem 12 transfers this bound to every centered real log-concave X . Since $\mathbb{E}\Gamma^{2m} = (2m)$ and $\mathbb{E}\Gamma^4 = 4 = 9$, the displayed constant follows. Taking $X = \Gamma$ shows sharpness. $\square$

5 Verification and limitations

The accompanying script `code/verify_p4_coefficients.py` derives the moments from (1) with exact rational arithmetic, checks (5), and verifies the positive coefficients in (4) through a selectable finite order. The command

```
conda run --no-capture-output -n sandbox python
code/verify_p4_coefficients.py --max-m 60
```

passes. This finite calculation is a regression test only; the proof is the uniform certificate in Lemma 1.

The full conjecture for even $p > 4$, or for arbitrary even $p < q$, remains open. The proof establishes strictness inside the reduced family away from the endpoints. It does not independently prove that equality in Eitan’s reduction forces the original X to equal the reduced extremizer, so the general equality classification is intentionally not claimed here.

A bounded novelty search checked Eitan’s paper, Murawski’s 2022 comparison paper [2], the final 2026 version of Melbourne–Roysdon–Tang–Tkocz [3], the 2026 exponential-sum paper of Brazitikos–Tang–Tkocz [4], and arXiv phrase searches for the $p = 4$ subcase. No prior all-even- q theorem was found. In particular, [3] still records the endpoint conjecture as known for $p = 2$ and for even $p, q \leq 100$. This is bounded evidence, not an exhaustive priority claim.

Human-review focus. Check the coefficient change from (8) to (11), the four-low-coefficient contribution (15), and the exact scope of Eitan’s Theorem 12.

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